

Function Call and Parameter Passing in C

A set of 5 example programs, from simplest to more advanced.

Program 1: Simplest function call (no parameters, no return)

This is the most basic function — no input, no output value, it just performs an action.

```
#include <stdio.h>

void welcome()
{
    printf("Welcome to C Programming!\n");
}

int main()
{
    welcome();    // function call
    return 0;
}
```

Output:

Welcome to C Programming!

Program 2: Function with one parameter (no return)

The same function can be called multiple times with different argument values.

```
#include <stdio.h>

void greet(char name[])
{
    printf("Hello, %s!\n", name);
}

int main()
{
    greet("Kumar");    // "Kumar" is passed as an argument
    greet("Priya");    // called again with a different argument
    return 0;
}
```

Output:

Hello, Kumar!
Hello, Priya!

Program 3: Function with parameters AND a return value

Parameters bring values IN, return sends a value OUT.

```
#include <stdio.h>

int add(int a, int b)
{
    int sum = a + b;
```

```

        return sum;
    }

    int main()
    {
        int result = add(10, 20);
        printf("Sum = %d\n", result);
        return 0;
    }

```

Output:

Sum = 30

Program 4: Multiple parameters of different types

A function can take several parameters of different types together.

```

#include <stdio.h>

void showStudent(char name[], int age, float marks)
{
    printf("Name: %s, Age: %d, Marks: %.2f\n", name, age, marks);
}

int main()
{
    showStudent("Ravi", 19, 88.5);
    return 0;
}

```

Output:

Name: Ravi, Age: 19, Marks: 88.50

Program 5: One function calling another function

Functions are not limited to being called only from main() — any function can call another.

```

#include <stdio.h>

int square(int x)
{
    return x * x;
}

int sumOfSquares(int a, int b)
{
    int result = square(a) + square(b);    // square() called here
    return result;
}

int main()
{
    int answer = sumOfSquares(3, 4);
    printf("Sum of squares = %d\n", answer);
    return 0;
}

```

Output:

Sum of squares = 25